

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical - IX

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Class: TYIT	Batch: 1
Date of Assignment: 18/08/2026	Date/Time of Submission: 18/08/2026

Student.cs

```
namespace practical_9A.Models
{
    public class Student
    {
        public int Id { get; set; }
        public string Name { get; set; }
        public int Age { get; set; }
        public string Course { get; set; }
    }
}
```

AppDbContext.cs

```
using Microsoft.EntityFrameworkCore;
using practical_9A.Models;
namespace StudentAPI.Data
{
    public class AppDbContext : DbContext
    {
        public AppDbContext(DbContextOptions<AppDbContext> options)
            : base(options)
        {
        }

        public DbSet<Student> Students { get; set; }
    }
}
```

appsettings.json

```
{
  "Logging": {
    "LogLevel": {
      "Default": "Information",
      "Microsoft.AspNetCore": "Warning"
    }
  },
  "AllowedHosts": "*",
}
```

```
"ConnectionStrings": {  
  "DefaultConnection": "Server=(localdb)\\MSSQLLocalDB;Database=StudentDB;Trusted_Connection=True;"  
}
```

Program.cs

```
using Microsoft.EntityFrameworkCore;  
using StudentAPI.Data;  
  
var builder = WebApplication.CreateBuilder(args);  
  
builder.Services.AddControllers();  
  
builder.Services.AddDbContext<AppDbContext>(options =>  
  options.UseSqlServer(  
    builder.Configuration.GetConnectionString("DefaultConnection")));  
  
builder.Services.AddEndpointsApiExplorer();  
builder.Services.AddSwaggerGen();  
  
var app = builder.Build();  
  
app.UseSwagger();  
app.UseSwaggerUI();  
  
app.UseHttpsRedirection();  
app.UseAuthorization();  
app.MapControllers();  
  
app.Run();
```

StudentsController.cs

```
using Microsoft.AspNetCore.Mvc;  
using practical_9A.Models;  
using StudentAPI.Data;  
  
namespace StudentAPI.Controllers  
{  
  [Route("api/[controller]")]  
  [ApiController]  
  public class StudentsController : ControllerBase  
  {  
    private readonly AppDbContext _context;  
  
    public StudentsController(AppDbContext context)  
    {  
      _context = context;  
    }  
  }  
}
```

```
// GET ALL
[HttpGet]
public IActionResult GetStudents()
{
    return Ok(_context.Students.ToList());
}

// GET BY ID
[HttpGet("{id}")]
public IActionResult GetStudent(int id)
{
    var student = _context.Students.Find(id);

    if (student == null)
        return NotFound();
    return Ok(student);
}

// INSERT
[HttpPost]
public IActionResult AddStudent(Student student)
{
    _context.Students.Add(student);
    _context.SaveChanges();

    return Ok(student);
}

// UPDATE
[HttpPut("{id}")]
public IActionResult UpdateStudent(int id, Student student)
{
    var data = _context.Students.Find(id);

    if (data == null)
        return NotFound();

    data.Name = student.Name;
    data.Age = student.Age;
    data.Course = student.Course;

    _context.SaveChanges();

    return Ok(data);
}
```

```
// DELETE
[HttpDelete("{id}")]
public IActionResult DeleteStudent(int id)
{
    var student = _context.Students.Find(id);

    if (student == null)
        return NotFound();
    _context.Students.Remove(student);
    _context.SaveChanges();

    return Ok();
}
}
```

